

KRISHNANUNNI C G

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EDUCATION

The University of Texas at Austin, TX

January 2021 – May 2026

Ph.D. in Aerospace Engineering & Engineering Mechanics (GPA: 3.81 / 4.0)

Indian Institute of Technology Madras, India

August 2017 – December 2019

Master of Science in Structural Engineering (GPA: 9.41 / 10)

National Institute of Technology Calicut, India

August 2013 – August 2017

Bachelor of Science in Civil Engineering (GPA: 9.15 / 10)

FELLOWSHIPS, SCHOLARSHIPS, and AWARDS

- Best poster award at the Workshop on Scientific Machine Learning, UT Austin. **October 2024**
- Warren A. and Alice L. Meyer Endowed Scholarship in Engineering from the Cockrell School of Engineering, UT Austin. **June 2024**
- Travel Awards by the United States Association for Computational Mechanics (June 2024), Society for Industrial and Applied Mathematics Texas-Louisiana (November 2022).
- Scholarship by the American Society of Indian Engineers and Architects (ASIE). **November 2022**
- Best MS Thesis award, Indian Institute of Technology Madras. **August 2020**
- Best Major B. Tech project award, National Institute of Technology, Calicut, India. **August 2017**
- Summer research fellowship, Department of mathematics, IISc, Indian Academy of Sciences. **July 2015**

JOURNAL PUBLICATIONS

- C. G. Krishnanunni., Tan Bui-Thanh., Clint Dawson (2025). Topological derivative approach for deep neural network architecture adaptation. ([Link](#)), Manuscript accepted at [SIAM Journal on Scientific Computing](#).
Contributions: Method development, theoretical analysis, and numerical implementation.
- C. G. Krishnanunni., Jonathan Wittmer., Tan Bui-Thanh., Quoc P. Nguyen (2026). A New Look at the Ensemble Kalman Filter for Inverse Problems: Duality, Non-Asymptotic Analysis and Convergence Acceleration. ([Link](#)) **Contributions: Method development, theoretical analysis, and numerical implementation.**
- William Cole Nockolds., C. G. Krishnanunni., Tan Bui-Thanh., Xianzhu Tang (2025). Lilan: A linear latent network approach for real-time solutions of stiff, nonlinear, ordinary differential equations. ([Link](#))
Contributions: Participation in method development and conducting the entire theoretical analysis.
- C. G. Krishnanunni., Tan Bui-Thanh (2025). An adaptive and stability-promoting layerwise training approach for sparse deep neural network architecture. *Computer Methods in Applied Mechanics and Engineering* ([Link](#))
Contributions: Method development, theoretical analysis, and numerical implementation.
- Jonathan Wittmer., C. G. Krishnanunni., Hai Van Nguyen., Tan Bui-Thanh (2023). On Unifying Randomized Methods for Inverse Problems. *Inverse Problems*. ([Link](#)) **Contributions: Participation in theoretical analysis and numerical implementation.**
- KS Suraj., PM Anilkumar., C. G. Krishnanunni., BN Rao (2023). Uncertainty quantification of bistable variable stiffness laminate using machine learning assisted perturbation approach. *Composite Structures* (accepted) ([Link](#)) **Contributions: Participation in method development and theoretical analysis.**
- Albert Orwa Akuno., L. Leticia Ramirez-Ramirez., Chahak Mehta., C. G. Krishnanunni., Tan Bui-Thanh., Jose Arturo Montoya (2022). Multi-patch epidemic models with partial mobility, residency, and demography. *Chaos, Solitons, & Fractals*. **Contributions: Participation in theoretical analysis** ([Link](#))
- C. G. Krishnanunni., B. N. Rao., (2021). Indirect health monitoring of bridges using Tikhonov regularization scheme and signal averaging technique. *Structural Control and Health Monitoring*. ([Link](#)) **Contributions: Method development, conducting the entire theoretical analysis, and numerical implementation**

Additional publications are available [HERE](#)

RECENT INVITED TALKS

- Topological derivative approach for deep neural network architecture adaptation. *Computational Algebra Seminar*, Department of Mathematics, UT Austin, May 2, 2025.
- Layerwise sparsifying training and sequential learning strategy for neural architecture adaptation. *U. S. National Congress on Computational Mechanics*, New Mexico, July 23-27, 2023.
- A two-stage strategy for neural architecture adaptation. *5th Annual meeting of the SIAM Texas-Louisiana Section on Uncertainty Quantification*, Houston, November 4-6, 2022.

RECENT RESEARCH EXPERIENCES

Solving forward and inverse problems in Plasma fusion research

Collaborator: Dr. Tan Bui-Thanh (UT Austin) & Dr. Xianzhu Tang (Los Alamos National Laboratory)

- Research aimed at i) adaptive strategies for learning a surrogate for the Collisional-Radiative (CR) model; and ii) solving an inverse problem for runaway electron mitigation during a plasma disruption event using evolutionary algorithms.

Transformer-powered generative model for solving inverse problems via joint modeling with forward process

Collaborator: Dr. Kowshik Thopalli, Dr. Yamen Mubarka, Dr. Vivek Narayanaswamy, Dr. Jayaraman J. Thiagarajan (Lawrence Livermore National Laboratory, USA)

- Designed a transformer architecture based generative model that transports samples from a prior distribution to samples from posterior parameter distribution conditioned on an input measurement.

Developing efficient algorithms for neural architecture adaptation

Collaborator: Dr. Tan Bui-Thanh (UT Austin, USA)

- Research in mathematical optimization and machine learning aimed at developing a mathematically principled way for automatically determining neural network architecture for a given data-set.

A new look at the Ensemble Kalman filter via duality

Collaborator: Dr. Tan Bui-Thanh (UT Austin, USA)

- Research aimed at the analysis of Ensemble Kalman filter for inverse problems in order to get insights into new convergence improvement strategies.

Mathematical epidemiology project

Collaborator: Dr. Tan Bui-Thanh (UT Austin, USA) & Leticia Ramirez-Ramirez (CIMAT, Mexico)

- Research aimed at developing an epidemic model that takes into account the effects of human mobility on the evolution of disease dynamics in a multi-population environment.

LEADERSHIP/MENTORSHIP ROLES

• Moncrief Summer Internship mentor:

- * Mentored [Jennifer Zheng](#) on a project titled *Physics informed deep-learning approach enhanced by POD for forecasting solutions to time-dependent PDE*.
- * Mentored [Giancarlo Villatoro](#) on a project titled [From an Elementary Proof of Error Representation for Hermite Quadrature to a Rediscovery of Legendre Polynomials and Rodrigues Formula](#)

- **SIAM-UT Mentorship program:** Worked with [Venkata Hasith Vattikuti](#) on a project focused on the use of reinforcement learning for solving a combinatorial optimization problem (nonlinear dimension reduction).

- **Roles in Conferences:** Co-organized mini-symposiums at [USNCCM17](#), [5th SIAM TX-LA Annual Meeting](#), [SIAM Conference on Mathematics of Data Science](#), Member of the local organizing committee for [8th SIAM TX-LA Annual Meeting](#) held at UT Austin.

PROFESSIONAL EXPERIENCE

- Graduate Teaching/Research Assistant** **January 2021 - Present**
Oden Institute of Computational Engineering & Sciences, UT Austin *Austin, TX*
- Research assistant to Prof. Tan Bui-Thanh, Institute of Computational Engineering and Sciences.
 - Teaching assistant for courses: Analytical methods, Mathematical methods in Science and Engineering.
- Computing graduate student intern** **June 2024 - August 2024**
Lawrence Livermore National Laboratory *Livermore, California*
- Research intern at the [Machine Intelligence Group](#), LLNL.
- Graduate Teaching/Research Assistant** **August 2017 - December 2020**
Indian Institute of Technology Madras *Madras, India*
- Research assistant to Prof. B. N. Rao, Structural Engineering department, IIT Madras.
 - Teaching assistant for courses: Structural optimization and Finite element analysis.

JOURNAL ROLES

Peer Reviewer: *Applied Mathematical Modelling*, Elsevier, *Sound and Vibration*, Elsevier, *Applied Ocean Research*, Elsevier.

SKILLS

Software: MATLAB[®], L^AT_EX[®], AutoCAD[®], ORIGIN[®], ANSYS [®]

Programming Languages: C++, Java, Python

ML Library: TensorFlow, PyTorch

REFERENCES

- **Tan Bui-Thanh**
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Leader of Pho-Ices group
Department of Aerospace Engineering and Engineering Mechanics
The Oden Institute for Computational Engineering and Sciences
The University of Texas at Austin
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- **Jayaraman J. Thiagarajan**
Principle Applied Scientist
AGI Science at Amazon
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- **Xianzhu Tang**
Physicist
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- **Kentaro Yaji**
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